

Polynomial-time approximation of justified envy-free Pareto-optimal lotteries in two-sided markets

W4414541699_p0 (website page)

Website #31 | Solver verdict: PARTIAL | Verifier verdict: n/a | Shown: solver output

Problem

We are given a bipartite market with sides A and B , $|A| = |B| = n$, nonnegative rational utilities

$$u_{ij} \quad (i \in A, j \in B), \quad w_{ji} \quad (j \in B, i \in A).$$

A feasible allocation is a fractional perfect matching

$$x = (x_{ij})_{i \in A, j \in B}, \quad x_{ij} \geq 0, \quad \sum_j x_{ij} = 1 \quad \forall i, \quad \sum_i x_{ij} = 1 \quad \forall j.$$

Utilities are

$$U_i(x) = \sum_j u_{ij} x_{ij}, \quad W_j(x) = \sum_i w_{ji} x_{ij}.$$

For $i, i' \in A$, define the A -side justified-envy benchmark

$$J_{ii'}(x) := \sum_{j \in B: w_{ji} \geq w_{ji'}} u_{ij} x_{ij},$$

and symmetrically, for $j, j' \in B$,

$$K_{jj'}(x) := \sum_{i \in A: u_{ij} \geq u_{ij'}} w_{ji} x_{ij}.$$

I adopt the strongest natural multiplicative formalization of approximate JEF:

x is α -approximately JEF if for all $i, i' \in A$ and $j, j' \in B$,

$$U_i(x) \geq \frac{1}{\alpha} J_{ii'}(x), \quad W_j(x) \geq \frac{1}{\alpha} K_{jj'}(x).$$

The target question is whether there is a universal constant $\alpha \geq 1$, independent of n , for which one can compute in polynomial time a feasible allocation that is simultaneously Pareto-optimal and α -approximately JEF.

The negation is: for every constant $\alpha \geq 1$, there exists an input instance for which no polynomial-time method can always output an allocation that is both Pareto-optimal and α -approximately JEF (equivalently, the existential version fails on some instance).

Alternative readings considered

The only real ambiguity is the formalization of “ α -approximately JEF.” I choose the pairwise multiplicative relaxation

$$U_i(x) \geq \alpha^{-1} J_{ii'}(x), \quad W_j(x) \geq \alpha^{-1} K_{jj'}(x),$$

because it is the closest multiplicative analogue of exact JEF and is stronger than looser formulations such as bounding only the maximum envy violation.

I treat outputting the allocation matrix x itself as sufficient, since the problem statement already identifies such an x with a lottery over integral perfect matchings.

Related literature found

- **Tröbst–Vazirani (2025)** — PARTIAL. They introduce JEF in this exact two-sided cardinal model, prove existence of JEF + weak Pareto-optimal allocations, show that Nash bargaining can exhibit $\Theta(n)$ -factor justified envy in two-sided markets, and explicitly list the constant-factor JEF + PO question as open.
- **Manjunath (2016)** — BACKGROUND. Introduces double-indexed-price equilibria for fractional matching markets; this is the equilibrium tool used by Tröbst–Vazirani for their JEF + weak-PO existence theorem, but I do not invoke it in the proof below.
- **Caragiannis–Filos–Ratsikas–Kanellopoulos–Vaish (2021)** — BACKGROUND. Studies cardinal *stable fractional matchings* and approximate stability/welfare tradeoffs. This is nearby but not identical: their pairwise blocking constraints are not the same as the aggregate JEF constraints used here.
- **He–Miralles–Yan (2026 working paper)** — PARTIAL. Gives justified-envy-free and ex ante efficient random assignments in a restricted many-to-one model where one side has cardinal preferences and the other side has only ordinal priorities. This does not cover the fully cardinal one-to-one model of the present problem, but it suggests that exact fairness/efficiency becomes easier once one side is weakened from cardinal to ordinal.

In the web searches I ran, I did not find a later paper resolving the exact general question above; the strongest directly relevant general result I found remained Tröbst–Vazirani’s JEF + weak-PO existence theorem and their explicit open question about constant-factor JEF + PO.

Proof sketch

A direct attack on multiplicative JEF is difficult because the JEF benchmarks $J_{ii'}(x)$ and $K_{jj'}(x)$ themselves depend on the output allocation x . Worse, Pareto-improving an allocation can increase these benchmarks, so a naive “find a fair point, then Pareto-clean it” step need not preserve direct JEF.

The key simplification is to use a *monotone certificate* for JEF: if every $i \in A$ receives at least a t -fraction of her personal maximum attainable utility

$$m_i := \max_{j \in B} u_{ij},$$

and every $j \in B$ receives at least a t -fraction of her personal maximum

$$M_j := \max_{i \in A} w_{ji},$$

then x is automatically $1/t$ -approximately JEF. The reason is elementary: any justified-envy benchmark for i is at most m_i , because it is a convex combination of the numbers u_{ij} ; similarly any justified-envy benchmark for j is at most M_j .

So the algorithm is:

1. maximize the largest common normalized floor t such that every A -agent gets at least tm_i and every B -agent gets at least tM_j ;
2. then Pareto-clean that solution while preserving all these floors.

Both steps are linear programs. The second LP outputs a genuinely Pareto-optimal allocation because any Pareto improvement would strictly increase the positive objective $\sum_i U_i + \sum_j W_j$ while respecting the preserved utility floors.

Finally, the uniform allocation $x_{ij} = 1/n$ certifies $t \geq 1/n$, so the worst-case approximation factor is at most n . This does *not* settle the constant- α question, but it gives a fully rigorous polynomial-time PO + multiplicative-JEF approximation guarantee for the general model.

Main result

Theorem 1. *Let*

$$m_i := \max_{j \in B} u_{ij} \quad (i \in A), \quad M_j := \max_{i \in A} w_{ji} \quad (j \in B).$$

Consider the linear program

$$\begin{aligned} & \text{maximize} && t \\ & \text{subject to} && \sum_{j \in B} x_{ij} = 1 \quad \forall i \in A, \\ & && \sum_{i \in A} x_{ij} = 1 \quad \forall j \in B, \\ & && x_{ij} \geq 0 \quad \forall i \in A, j \in B, \\ & && U_i(x) \geq t m_i \quad \forall i \in A, \\ & && W_j(x) \geq t M_j \quad \forall j \in B, \\ & && 0 \leq t \leq 1. \end{aligned} \tag{LP-Floor}$$

Let t^ be its optimum, and let x^{fl} be an optimal allocation.*

Now solve

$$\begin{aligned} & \text{maximize} && \sum_{i \in A} U_i(y) + \sum_{j \in B} W_j(y) \\ & \text{subject to} && \sum_{j \in B} y_{ij} = 1 \quad \forall i \in A, \\ & && \sum_{i \in A} y_{ij} = 1 \quad \forall j \in B, \\ & && y_{ij} \geq 0 \quad \forall i \in A, j \in B, \\ & && U_i(y) \geq U_i(x^{\text{fl}}) \quad \forall i \in A, \\ & && W_j(y) \geq W_j(x^{\text{fl}}) \quad \forall j \in B. \end{aligned} \tag{LP-PO}$$

Return any optimum y of (LP-PO).

Then y is Pareto-optimal and α^ -approximately JEF with*

$$\alpha^* = \frac{1}{t^*}.$$

Moreover $t^ \geq \frac{1}{n}$, so $\alpha^* \leq n$.*

In particular, there is a polynomial-time algorithm that always outputs a Pareto-optimal n -approximately JEF allocation.

Proof

I split the proof into three lemmas.

Lemma 1. *(LP-Floor) is always feasible, and its optimum satisfies $t^* \geq 1/n$.*

Proof. Consider the uniform allocation $\bar{x}$ defined by

$$\bar{x}_{ij} = \frac{1}{n} \quad \forall i \in A, j \in B.$$

It is feasible because for each i ,

$$\sum_{j \in B} \bar{x}_{ij} = n \cdot \frac{1}{n} = 1,$$

and for each j ,

$$\sum_{i \in A} \bar{x}_{ij} = n \cdot \frac{1}{n} = 1.$$

Now fix $i \in A$. Since $m_i = \max_j u_{ij}$, one term in the sum $\sum_j u_{ij}$ equals m_i and all terms are nonnegative. Therefore

$$U_i(\bar{x}) = \sum_{j \in B} u_{ij} \frac{1}{n} = \frac{1}{n} \sum_{j \in B} u_{ij} \geq \frac{m_i}{n}.$$

The same argument on the B -side gives, for every $j \in B$,

$$W_j(\bar{x}) = \sum_{i \in A} w_{ji} \frac{1}{n} = \frac{1}{n} \sum_{i \in A} w_{ji} \geq \frac{M_j}{n}.$$

Hence $(\bar{x}, 1/n)$ is feasible for (LP-Floor), so $t^* \geq 1/n$. □

Lemma 2. *Let x be any feasible allocation, and let $t \in [0, 1]$ satisfy*

$$U_i(x) \geq t m_i \quad \forall i \in A, \quad W_j(x) \geq t M_j \quad \forall j \in B.$$

Then x is $1/t$ -approximately JEF.

Proof. Fix $i, i' \in A$. By definition,

$$J_{ii'}(x) = \sum_{j: w_{ji} \geq w_{ji'}} u_{ij} x_{i'j}.$$

Since $u_{ij} \leq m_i$ for every j , we have

$$J_{ii'}(x) \leq \sum_{j: w_{ji} \geq w_{ji'}} m_i x_{i'j} \leq m_i \sum_{j \in B} x_{i'j} = m_i.$$

Using the hypothesis $U_i(x) \geq t m_i$, it follows that

$$U_i(x) \geq t m_i \geq t J_{ii'}(x),$$

or equivalently,

$$U_i(x) \geq \frac{1}{1/t} J_{ii'}(x).$$

Now fix $j, j' \in B$. Similarly,

$$K_{jj'}(x) = \sum_{i: u_{ij} \geq u_{ij'}} w_{ji} x_{ij'}.$$

Since $w_{ji} \leq M_j$ for every i ,

$$K_{jj'}(x) \leq \sum_{i: u_{ij} \geq u_{ij'}} M_j x_{ij'} \leq M_j \sum_{i \in A} x_{ij'} = M_j.$$

Using $W_j(x) \geq t M_j$, we get

$$W_j(x) \geq t M_j \geq t K_{jj'}(x),$$

which is the B -side $1/t$ -approximate JEF inequality.

Since the choice of i, i', j, j' was arbitrary, x is $1/t$ -approximately JEF. $\square$

Lemma 3. *Any optimal solution y of (LP-PO) is Pareto-optimal.*

Proof. Assume for contradiction that y is not Pareto-optimal. Then there exists another feasible allocation z such that

$$U_i(z) \geq U_i(y) \quad \forall i \in A, \quad W_j(z) \geq W_j(y) \quad \forall j \in B,$$

and at least one of these inequalities is strict.

Because y is feasible for (LP-PO), it satisfies

$$U_i(y) \geq U_i(x^{\text{fl}}) \quad \forall i \in A, \quad W_j(y) \geq W_j(x^{\text{fl}}) \quad \forall j \in B.$$

Combining these inequalities with the Pareto domination of y by z , we obtain

$$U_i(z) \geq U_i(x^{\text{fl}}) \quad \forall i \in A, \quad W_j(z) \geq W_j(x^{\text{fl}}) \quad \forall j \in B.$$

Hence z is also feasible for (LP-PO).

Moreover,

$$\sum_{i \in A} U_i(z) + \sum_{j \in B} W_j(z) > \sum_{i \in A} U_i(y) + \sum_{j \in B} W_j(y),$$

because every term on the left is at least the corresponding term on the right and at least one is strictly larger. This contradicts the optimality of y in (LP-PO).

Therefore y is Pareto-optimal. $\square$

We now prove the theorem.

Proof of the theorem. Let (x^{fl}, t^*) be an optimum of (LP-Floor), and let y be an optimum of (LP-PO).

By Lemma 1, $t^* \geq 1/n$.

By feasibility of y in (LP-PO),

$$U_i(y) \geq U_i(x^{\text{fl}}) \geq t^* m_i \quad \forall i \in A,$$

and

$$W_j(y) \geq W_j(x^{\text{fl}}) \geq t^* M_j \quad \forall j \in B.$$

Applying Lemma 2 with $t = t^*$, we conclude that y is $(1/t^*)$ -approximately JEF.

Applying Lemma 3, we conclude that y is Pareto-optimal.

Thus y is Pareto-optimal and α^* -approximately JEF for

$$\alpha^* = \frac{1}{t^*}.$$

Since $t^* \geq 1/n$, we get $\alpha^* \leq n$.

Finally, both (LP-Floor) and (LP-PO) are rational linear programs of polynomial size in n and the input encoding length, so they are solvable in polynomial time by standard polynomial-time algorithms for linear programming; see Invoked Result 1 below. $\square$

Gap to the full statement

The original problem asks for a *constant* α , independent of n . The theorem above gives

$$\alpha^* \leq n,$$

and in fact computes the best instance-dependent factor obtainable from the common normalized-utility-floor certificate. This is a genuine polynomial-time PO + multiplicative-JEF approximation theorem for the full model, but it does *not* settle whether the worst-case bound can be improved to an absolute constant.

What remains open after this manuscript is exactly the question highlighted by Tröbst–Vazirani: can one replace the worst-case n by some universal constant, or prove that no such constant exists?

Self red-team

- **Counterexample attempt against Lemma 2 (floor $\Rightarrow$ approximate JEF).** I tried markets where the envied agent i' concentrates all mass on a single partner j that strongly prefers i . The benchmark $J_{ii'}(x)$ is then as large as possible. The lemma survives because $J_{ii'}(x)$ is still at most m_i , independently of how pathological the consent pattern is.
- **Counterexample attempt against Lemma 3 (Pareto cleanup).** I tried to exploit the fact that (LP-PO) optimizes only total welfare under floors. The worry is that some other feasible point might Pareto-dominate y without changing total welfare. This cannot happen: under Pareto domination at least one individual's utility is strictly larger and nobody's is smaller, so the total welfare objective is strictly larger.
- **All-zero and zero-row boundary cases.** If some $m_i = 0$, then $u_{ij} = 0$ for all j , so every A-side JEF benchmark involving i is also 0; the approximation claim is then vacuous and correct. If all $m_i = M_j = 0$, the cap $0 \leq t \leq 1$ in (LP-Floor) prevents unboundedness, and every feasible allocation is trivially exact JEF and PO iff it is any feasible allocation.
- **Most plausible referee objection #1.** “Your Pareto-cleanup step may destroy JEF, because JEF benchmarks depend on the final allocation.” Response: correct for *direct* JEF constraints; false for the utility-floor certificate. The proof deliberately uses the monotone certificate $U_i \geq tm_i$, $W_j \geq tM_j$, whose validity for $1/t$ -JEF is independent of other agents' final bundles.
- **Most plausible referee objection #2.** “Is the approximation notion you fixed too weak?” Response: I chose the strongest natural pairwise multiplicative analogue of exact JEF. Any weaker multiplicative reading would only strengthen the partial-result verdict.
- **Most plausible referee objection #3.** “Does the proof accidentally imply a constant factor?” Response: no. The only universal lower bound I prove is $t^* \geq 1/n$, coming from the uniform allocation. I do not derive any $t^* \geq c > 0$ independent of n .
- **Boundary checks.** For $n = 1$, (LP-Floor) gives $t^* = 1$, and the returned allocation is exact JEF and PO. For symmetric binary markets where exact EF + PO is known to exist, my algorithm remains correct but is generally not tight; this is expected, since the theorem is a worst-case approximation guarantee rather than an exact characterization.

Ideas tried

1. **Directly optimize fixed- t JEF constraints.** For fixed $t = 1/\alpha$, the inequalities

$$U_i(x) \geq t J_{ii'}(x), \quad W_j(x) \geq t K_{jj'}(x)$$

are linear in x , so one can optimize over the t -JEF polytope. *Obstruction:* maximizing welfare over that restricted polytope gives Pareto-optimality only *within* the t -JEF set, not globally. A subsequent Pareto-cleanup step can increase the JEF benchmarks and destroy t -JEF. *Next step:* one would need a way to certify global PO simultaneously with t -JEF, likely via a supported-efficient characterization that is still polynomially searchable.

2. **Upgrade JEF + weak PO to JEF/constant-JEF + PO via perturbation of supporting weights.** Tröbst–Vazirani prove existence of JEF + weak PO using a limiting equilibrium argument. *Obstruction:* weakly Pareto-optimal points can lie deep inside a supported face, and small positive perturbations of the supporting weights can jump to a far-away Pareto-optimal point. That move can badly change the JEF benchmarks. *Next step:* one would need a continuity theorem tailored to JEF faces, not just to utility faces.
3. **Use star-type impossibility gadgets inspired by the Nash-bargaining lower bound.** Theorem 46 of Tröbst–Vazirani shows that Nash bargaining itself can have $\Theta(n)$ -factor justified envy. *Obstruction:* those gadgets do not force *every* Pareto-optimal allocation to have large justified envy. Fractional balancing along the Pareto frontier can often keep the envy ratio constant, so the gadget disproves the mechanism, not the existential claim.
4. **Relate JEF to cardinal stable fractional matchings.** Stable fractional matchings satisfy pairwise blocking constraints of the form “for each pair (i, j) , either $U_i \geq u_{ij}$ or $W_j \geq w_{ji}$.” *Obstruction:* JEF aggregates over the *portion of another agent’s bundle* assigned to acceptable partners. Pairwise stability does not control this aggregate benchmark when a partner already obtains high utility from even better agents. *Next step:* a hybrid notion mixing pairwise stability with partner-specific exposure constraints might bridge the gap.

Invoked results

1. **Polynomial-time solvability of rational linear programming. Statement.** Linear programs with rational data can be solved in time polynomial in the encoding length of the input. **Classification.** T00L. **Citation.** Khachiyan (1979), *Doklady Akademii Nauk SSSR* 244(5):1093–1096. **Hypothesis audit.** Both (LP-Floor) and (LP-PO) have polynomially many variables and constraints: $O(n^2)$ allocation variables, plus one scalar t in the first LP. All coefficients are rational numbers drawn from the input utilities and the derived maxima m_i, M_j , which are themselves rational and computable in polynomial time. Therefore the theorem applies exactly as needed.

Self-confidence

4 — The proved result is elementary, the argument is fully written out, and after red-teaming I do not see a gap; the only unresolved part is the original constant- α objective, which this manuscript does not claim to settle.

Bibliography

```
@article{TrobstVazirani2025,
  author = {Thorben Trobst and Vijay V. Vazirani},
  title = {Cardinal-Utility Matching Markets: The Quest for Envy-Freeness, Pareto-Optimality, and
    Efficient Computability},
  journal = {Mathematics of Operations Research},
  year = {2025},
  doi = {10.1287/moor.2024.0770},
  note = {Published online September 26, 2025}
}

@article{Manjunath2016,
  author = {Vikram Manjunath},
  title = {Fractional Matching Markets},
  journal = {Games and Economic Behavior},
  volume = {100},
  pages = {321--336},
  year = {2016},
  doi = {10.1016/j.geb.2016.10.006}
}

@article{CaragiannisEtAl2021,
  author = {Ioannis Caragiannis and Aris Filos-Ratsikas and Panagiotis Kanellopoulos and Rohit
    Vaish},
  title = {Stable Fractional Matchings},
  journal = {Artificial Intelligence},
  volume = {295},
  pages = {103416},
  year = {2021},
  doi = {10.1016/j.artint.2020.103416}
}

@techreport{HeMirallesYan2026,
  author = {Yinghua He and Antonio Miralles and Jianye Yan},
  title = {Competitive Equilibrium from Equal Incomes for Two-Sided Matching},
  institution = {Barcelona School of Economics Working Paper 692},
  year = {2026},
  month = may
}

@article{Khachiyan1979,
  author = {L. G. Khachiyan},
  title = {A Polynomial Algorithm in Linear Programming},
  journal = {Doklady Akademii Nauk SSSR},
  volume = {244},
  number = {5},
  pages = {1093--1096},
  year = {1979}
}
```